Light

GCSE Computer Science - Networks Basics (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is a network protocol?

- ○ **A.** A set of rules for transmitting data across a network
- ○ **B.** A type of network cable used to connect devices
- ○ **C.** Software that protects a network from viruses
- ○ **D.** A device that stores network passwords

Q2 (1 mark)

What is HTTP used for?

- ○ **A.** Transferring web pages between servers and browsers

- ○ **B.** Sending and receiving emails
- ○ **C.** Encrypting files on a hard drive
- ○ **D.** Connecting to a wireless network

Q3 (1 mark)

What is the main difference between HTTP and HTTPS?

- ○ **A.** HTTPS is faster than HTTP
- ○ **B.** HTTPS encrypts data for secure transmission
- ○ **C.** HTTPS only works on mobile devices
- ○ **D.** HTTPS is used for email while HTTP is for web pages

Q4 (1 mark)

What is FTP used for?

- ○ **A.** Browsing web pages
- ○ **B.** Transferring files between computers over a network
- ○ **C.** Sending instant messages
- ○ **D.** Making voice calls over the Internet

Q5 (1 mark)

Which protocol is used for sending emails?

- A. HTTP
- B. FTP
- C. SMTP
- D. DNS

Q6 (1 mark)

Which protocols are used for receiving emails?

- A. POP and IMAP
- B. HTTP and HTTPS
- C. FTP and SFTP
- D. TCP and UDP

Q7 (1 mark)

What is encryption in computer networks?

- A. Converting data into a coded form to prevent unauthorised access
- B. Compressing data to make files smaller
- C. Speeding up data transmission across networks
- D. Backing up data to a secondary location

Q8 (1 mark)

Which of the following is a valid IPv4 address?

- ○ **A.** 192.168.1.1
- ○ **B.** 2001:0db8:85a3:0000
- ○ **C.** AA:BB:CC:DD:EE:FF
- ○ **D.** www.example.com

Q9 (1 mark)

Why was IPv6 developed to replace IPv4?

- ○ **A.** IPv6 is faster than IPv4
- ○ **B.** IPv4 addresses were running out due to limited address space
- ○ **C.** IPv6 uses simpler numbers that are easier to remember
- ○ **D.** IPv4 could not work with wireless networks

Q10 (1 mark)

What is a MAC address?

- ○ **A.** A unique hardware identifier permanently assigned to a network interface
- ○ **B.** A temporary address assigned when connecting to a network
- ○ **C.** The address of a website on the Internet

- ○ **D.** A password used to access a wireless network

Q11 (1 mark)

Why are network standards important?

- ○ **A.** They make networks run faster
- ○ **B.** They allow hardware and software from different manufacturers to work together
- ○ **C.** They reduce the cost of network equipment
- ○ **D.** They prevent all cyber attacks

Q12 (1 mark)

What is Bluetooth primarily used for?

- ○ **A.** Short-range wireless communication between devices
- ○ **B.** Connecting to the Internet from anywhere in the world
- ○ **C.** Transferring large files across long distances
- ○ **D.** Providing high-speed broadband connections

Q13 (2 marks)

TCP/IP is described as a 'layered' protocol model.
What is a key benefit of using layers in network protocols?

- ○ **A.** It makes networks physically smaller
- ○ **B.** It automatically encrypts all data
- ○ **C.** It eliminates the need for any network hardware
- ○ **D.** Each layer can be developed and updated independently without affecting other layers

Q14 (2 marks)

Sarah uses two devices to check her email: a laptop and a smartphone. Which email protocol would best keep her emails synchronised across both devices?

- ○ **A.** POP, because it downloads emails faster
- ○ **B.** SMTP, because it handles both sending and receiving
- ○ **C.** IMAP, because it keeps emails on the server and synchronises across devices
- ○ **D.** FTP, because it transfers files between multiple locations

Q15 (2 marks)

A user wants to connect wireless headphones to their smartphone for listening to music.
Which wireless technology would be most appropriate?

- ○ **A.** Wi-Fi, because it has a longer range
- ○ **B.** 4G, because it works anywhere with mobile signal
- ○ **C.** Ethernet, because it provides the most stable connection

○ **D.** Bluetooth, because it is designed for short-range device connections with low power consumption

Q16 (2 marks)

A customer enters their credit card details on an online shopping website.
Why is encryption essential for this transaction?

○ **A.** To make the website load faster

○ **B.** To allow the transaction to work on mobile devices

○ **C.** To store the credit card details permanently on the customer's computer

○ **D.** To prevent the credit card data from being read if intercepted during transmission

Q17 (2 marks)

What is a key difference between an IP address and a MAC address?

○ **A.** IP addresses are longer than MAC addresses

○ **B.** MAC addresses are assigned to hardware and are permanent, while IP addresses can change

○ **C.** IP addresses only work on wireless networks

○ **D.** MAC addresses are used for Internet communication, while IP addresses are for local networks only

Q18 (2 marks)

A gaming company needs the fastest and most reliable network connections for their office computers used in competitive gaming tournaments.
Which connection type should they choose?

- ○ **A.** Wi-Fi, because it allows players to move around freely
- ○ **B.** Bluetooth, because it has low latency
- ○ **C.** Ethernet, because wired connections provide faster speeds and lower latency than wireless
- ○ **D.** 4G mobile, because it works without any infrastructure

Q19 (2 marks)

A network engineer needs to update the software that handles how data packets are routed across the Internet.
How does using a layered protocol model help with this update?

- ○ **A.** The entire network must be rebuilt from scratch
- ○ **B.** The update is impossible because protocols cannot be changed
- ○ **C.** All connected devices must be replaced with new hardware
- ○ **D.** Only the routing layer needs to be updated, leaving other layers unchanged

Q20 (2 marks)

A user notices that a website's URL starts with 'http://' instead of 'https://' when they are about to enter their password.
What security concern should this raise?

- ○ **A.** The website will load more slowly
- ○ **B.** The password could be intercepted because the connection is not encrypted
- ○ **C.** The website is definitely a scam
- ○ **D.** The user's computer will get a virus

Q21 (2 marks)

An IPv4 address looks like 192.168.1.1, while an IPv6 address looks like 2001:0db8:85a3:0000:0000:8a2e:0370:7334.
What explains the difference in format?

- ○ **A.** IPv6 is designed to be easier for humans to read
- ○ **B.** IPv4 was designed for wireless networks while IPv6 is for wired networks
- ○ **C.** IPv6 uses 128 bits instead of 32 bits, requiring a longer format to represent more addresses
- ○ **D.** IPv6 only uses letters while IPv4 only uses numbers

Q22 (2 marks)

A network administrator sees the address 00:1A:2B:3C:4D:5E in their network logs.
What type of address is this?

- ○ **A.** An IPv4 address
- ○ **B.** An IPv6 address
- ○ **C.** A MAC address

- ○ **D.** A domain name

Q23 (2 marks)

A school purchases network equipment from three different manufacturers.
Why can they expect all this equipment to work together on the same network?

- ○ **A.** All network equipment is made in the same factory
- ○ **B.** Network standards ensure equipment from different manufacturers follows the same rules
- ○ **C.** The school must pay extra for compatibility software
- ○ **D.** Only the most expensive equipment will be compatible

Q24 (3 marks)

James sends an email from his work computer to his colleague Maria. Which sequence of protocols is used to deliver this email?

- ○ **A.** HTTP to send, FTP to deliver, DNS to receive
- ○ **B.** FTP to upload, HTTP to transfer, SMTP to download
- ○ **C.** SMTP to send from James's computer to the mail servers, then POP or IMAP for Maria to retrieve it
- ○ **D.** TCP to send, UDP to transfer, IP to receive

Q25 (3 marks)

A bank's website uses HTTPS for all customer transactions.

A hacker intercepts data being transmitted between a customer and the bank's server.
Why is the hacker unable to steal the customer's account details?

- ○ **A.** HTTPS blocks all hackers from accessing the network
- ○ **B.** HTTPS encrypts the data, so the intercepted data appears as meaningless code without the decryption key
- ○ **C.** HTTPS makes data invisible during transmission
- ○ **D.** HTTPS deletes sensitive data before transmission

Q26 (3 marks)

A network administrator is troubleshooting a problem where users cannot access a specific website, but all other websites work fine.
How does the layered protocol model help identify the problem?

- ○ **A.** The administrator must replace all network equipment to fix any problem
- ○ **B.** The administrator cannot troubleshoot because layers hide all problems
- ○ **C.** The layered model automatically fixes all network problems
- ○ **D.** The administrator can systematically check each layer to isolate where the problem occurs

Q27 (3 marks)

A library is setting up a new computer network. They need to provide:

- Fast, reliable connections for 10 staff desktop computers that never move
- Internet access for visitors using their own laptops and phones

Which combination of connection types should they use?

- ○ **A.** Wi-Fi for everything because it's more convenient
- ○ **B.** Ethernet for everything because it's more reliable
- ○ **C.** Ethernet for staff desktops, Wi-Fi for visitors
- ○ **D.** Bluetooth for staff desktops, 4G for visitors

Q28 (3 marks)

When a computer sends data across a local network, it uses both the destination IP address and the destination MAC address.
Why are both types of address needed?

- ○ **A.** They are interchangeable and either one would work alone
- ○ **B.** IP addresses identify the network location, while MAC addresses identify the specific physical device to deliver data to
- ○ **C.** MAC addresses are only used for wireless connections
- ○ **D.** IP addresses are only used for Internet connections, not local networks

Q29 (3 marks)

A company develops a new type of smart home device. They want it to connect to customers' existing home Wi-Fi networks.
Why must the company follow Wi-Fi standards when building their device?

- ○ **A.** Wi-Fi standards make the device cheaper to manufacture
- ○ **B.** The company can create their own wireless technology that will work better
- ○ **C.** Standards are only required for professional network equipment

○ **D.** Without following standards, the device would not be able to communicate with existing routers and access points

Q30 (3 marks)

TCP/IP is the foundation protocol suite of the Internet.
Which statement best describes how TCP and IP work together?

○ **A.** TCP and IP perform exactly the same function and are interchangeable

○ **B.** IP encrypts data for security while TCP compresses data for speed

○ **C.** TCP is used for wired networks and IP is used for wireless networks

○ **D.** IP handles addressing and routing packets across networks, while TCP ensures reliable delivery by managing connections and retransmitting lost packets

Total: **55** marks • Time: **30 min**

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